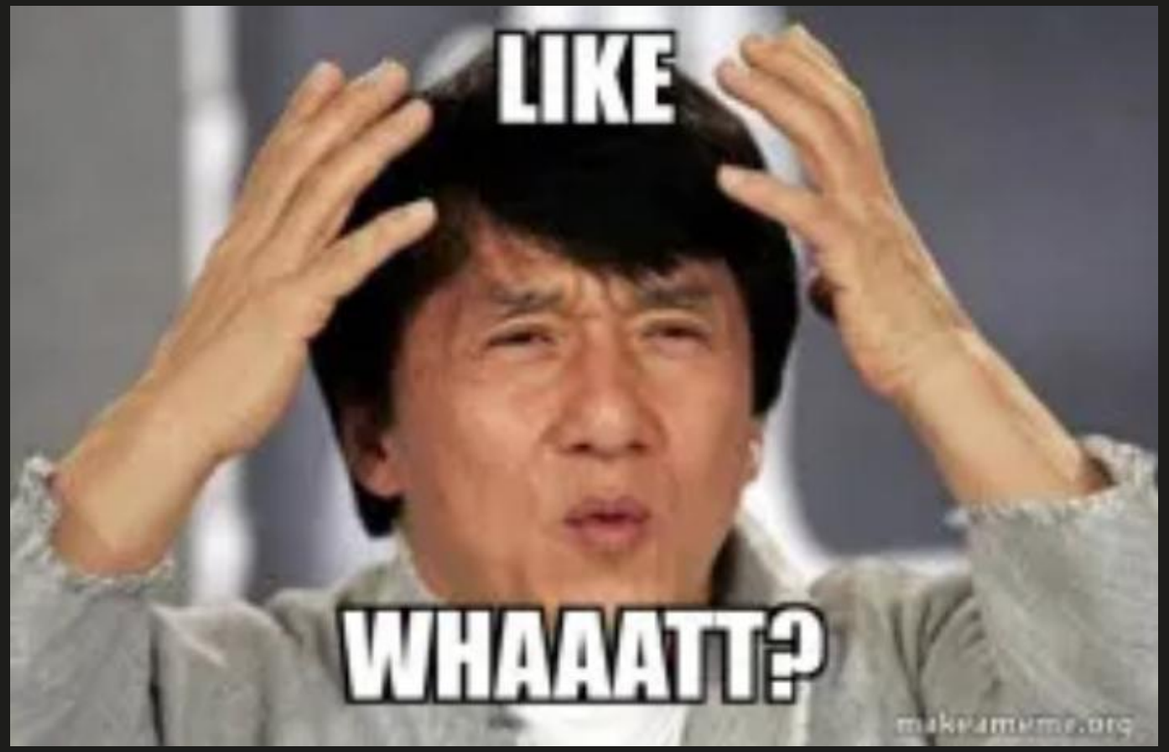
Sliding Contact Bearings: An Overview

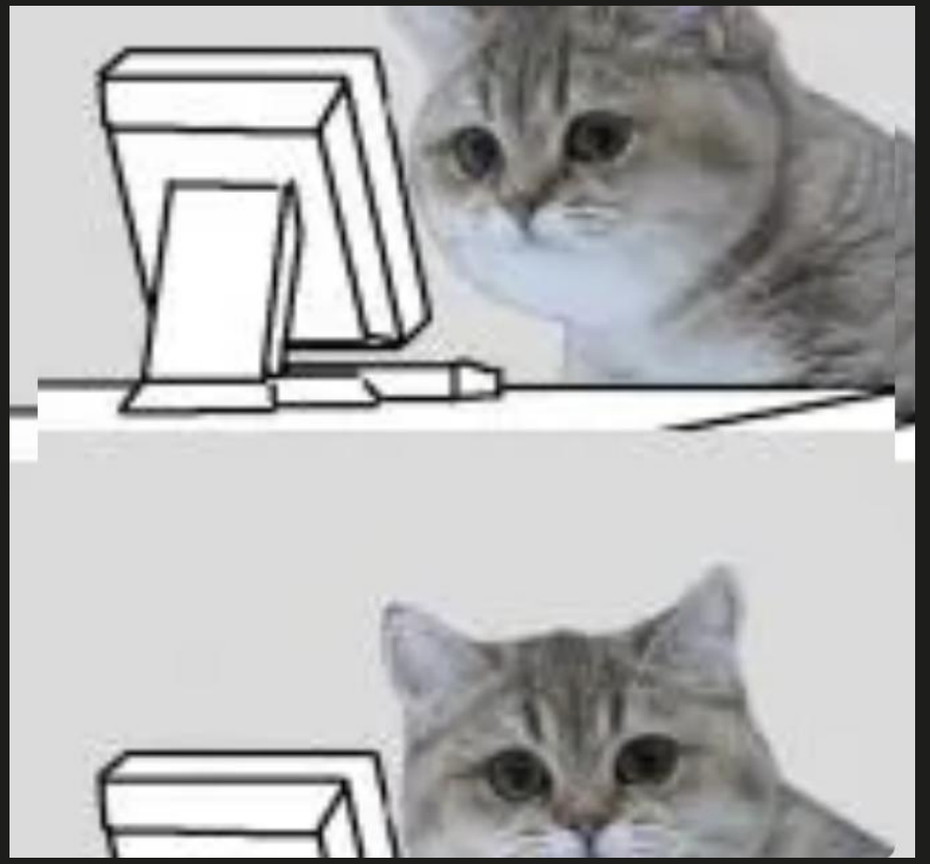
This presentation will explore the key aspects of sliding contact bearings, including their types, lubrication regimes, materials, advantages, disadvantages, applications, and future trends. Understanding these elements is crucial for selecting and maintaining bearings for optimal performance.



What does SPPU want?...

Unit 3	Sliding and Rolling Contact Bearing	07 Hrs.
Sliding contact bearing (Theoretical treatment only): Introduction to sliding contact bearing, classification, Reynolds's equation (2D), Petroff's equations, Sommerfeld number, Parameters of bearing design.		


LIKE
WHAAATT?



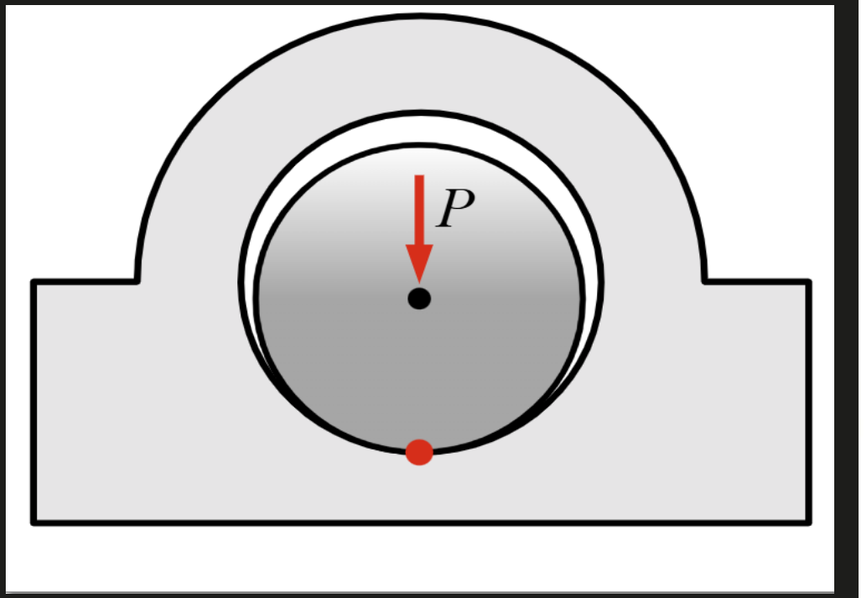
Types of Sliding Bearings

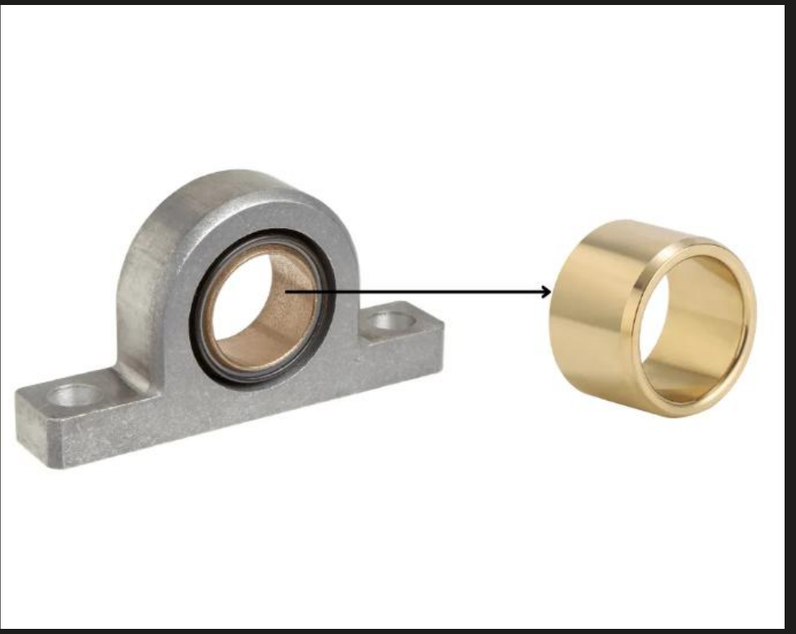
1. Journal Bearings

2.Thrust Bearings

3.Linear Bearings

Journal Bearings





Journal bearings are among the simplest types of sliding contact bearings. These bearings support a rotating shaft, often called a journal, within a cylindrical bearing surface. Lubrication is a critical aspect, where a fluid film generated by shaft rotation supports the load.

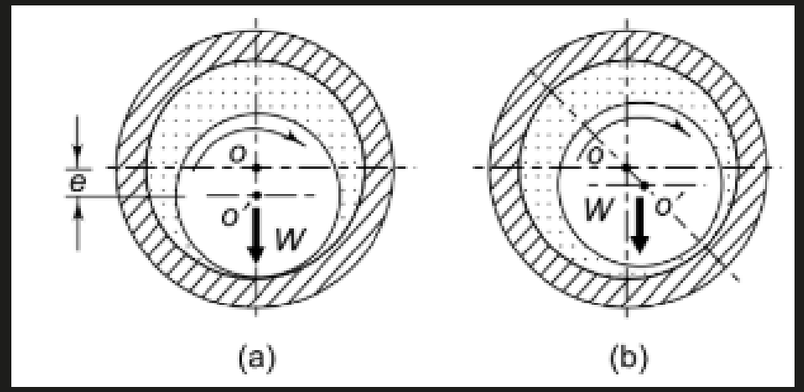
Lubrication Types

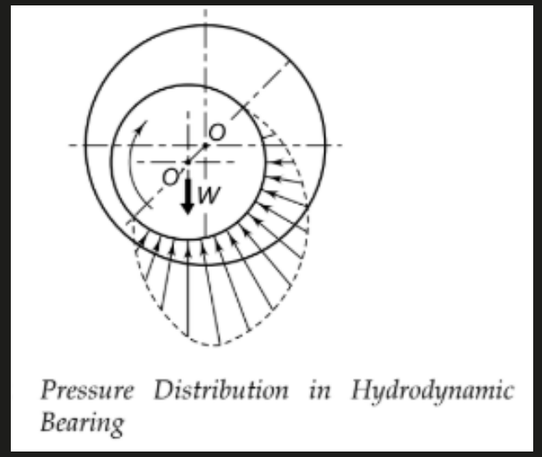
- Hydrodynamic lubrication: Self-generating film.
- Hydrostatic lubrication: External pump supplied.
- Thin film lubrication
- Zero film lubrication

Material Selection

- Babbitt alloys: Excellent conformability.
- Bronze: Good wear properties.
- Aluminum alloys: High load capacity.
- Polymers: Low friction.

Hydrodynamic Lubricating bearing:-





Pressure Distribution in Hydrodynamic Bearing

✓ No Metal-to-Metal Contact – Operates with a full film of lubricant.
✓ Self-Sustaining Lubrication – Requires no external pressure supply (unlike hydrostatic bearings).
✓ Load Carrying Capacity Increases with Speed – Higher speeds generate thicker oil films.
✓ Requires Precise Design – Clearance, viscosity, and speed must be optimized for effective operation.

Advantages

✓ High Load Capacity.

✓ Infinite Life with Proper Lubrication

✓ Quiet & Vibration-Free Operation

✓ Good Damping Properties

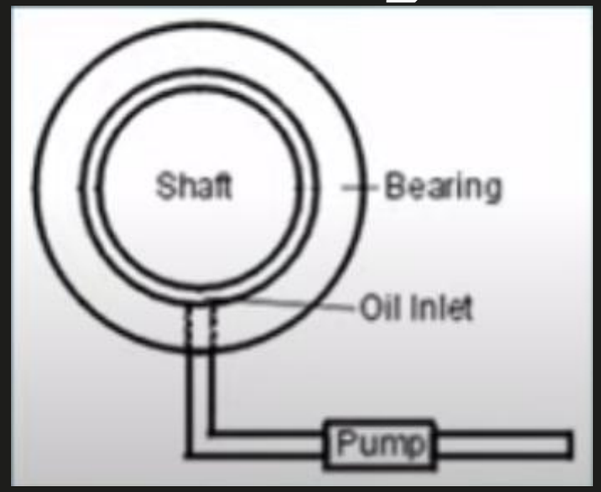
Disadvantages

✗ Requires Start-up Lubrication

✗ High Power Loss Due to Fluid Friction

✗ Overheating Risk

Hydrostatic lubrication Bearing:-



✓ External Lubrication Source – Uses a pump to deliver lubricant under pressure.
✓ No Wear at Start/Stop – Ideal for applications with frequent stops and starts.
✓ Very Low Friction & Heat Generation – Due to continuous oil film.
✓ High Precision & Stiffness – Used in machine tools and high-accuracy applications.

Advantages

✓ Supports Load at Rest

✓ Virtually No Wear

✓ Better Damping & Stability

✓ Suitable for Low & High Speeds

Disadvantages

✗ Requires an External Pump & Control System

✗ Energy Consumption

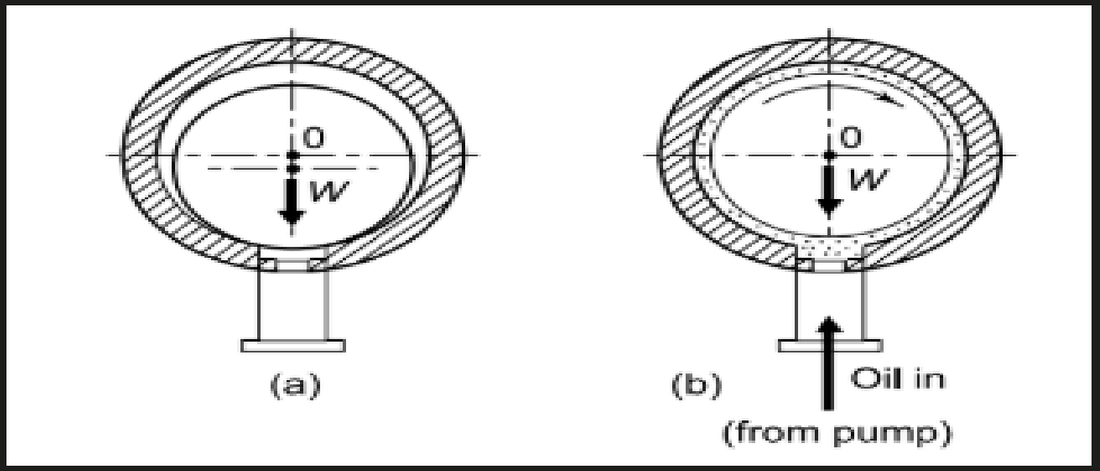
✗ Risk of Oil Supply Failure.

- Thin film lubrication:-





Hydro Dynamic	Hydro Static
No External Pressure Supply	External Power Supply
Gradual Pressure Development	Constant Pressure
High initial friction	No initial friction
Limited span of load carrying capacity	High range of load carrying capacity
Simple construction and low cost	Complex construction and high cost





Thrust Bearings

Thrust bearings are specifically designed to support ***axial loads*** along a shaft. There are two main types of thrust bearings. Flat thrust bearings have a simple design, suitable for low-speed applications. Tapered land thrust bearings feature grooved surfaces that promote hydrodynamic lubrication. These bearings find applications in gearboxes, turbines, and pumps.

Axial Load Support

Designed to handle forces parallel to the shaft's axis.

Types

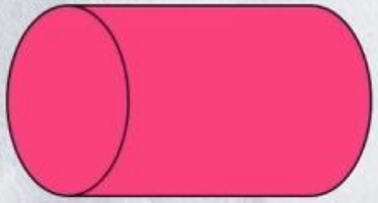
Flat and tapered designs.

Applications

Used in gearboxes, turbines, and pumps.



Ball



Roller



Needle



Thrust ball
Bearing

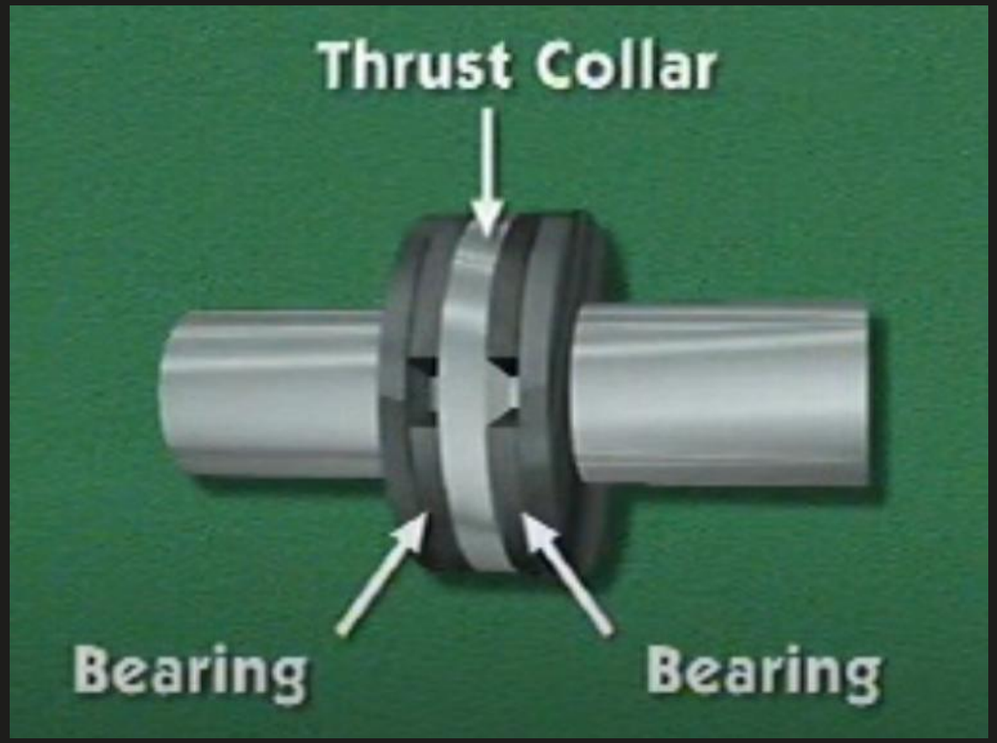


Cylindrical Roller
Thrust Bearing



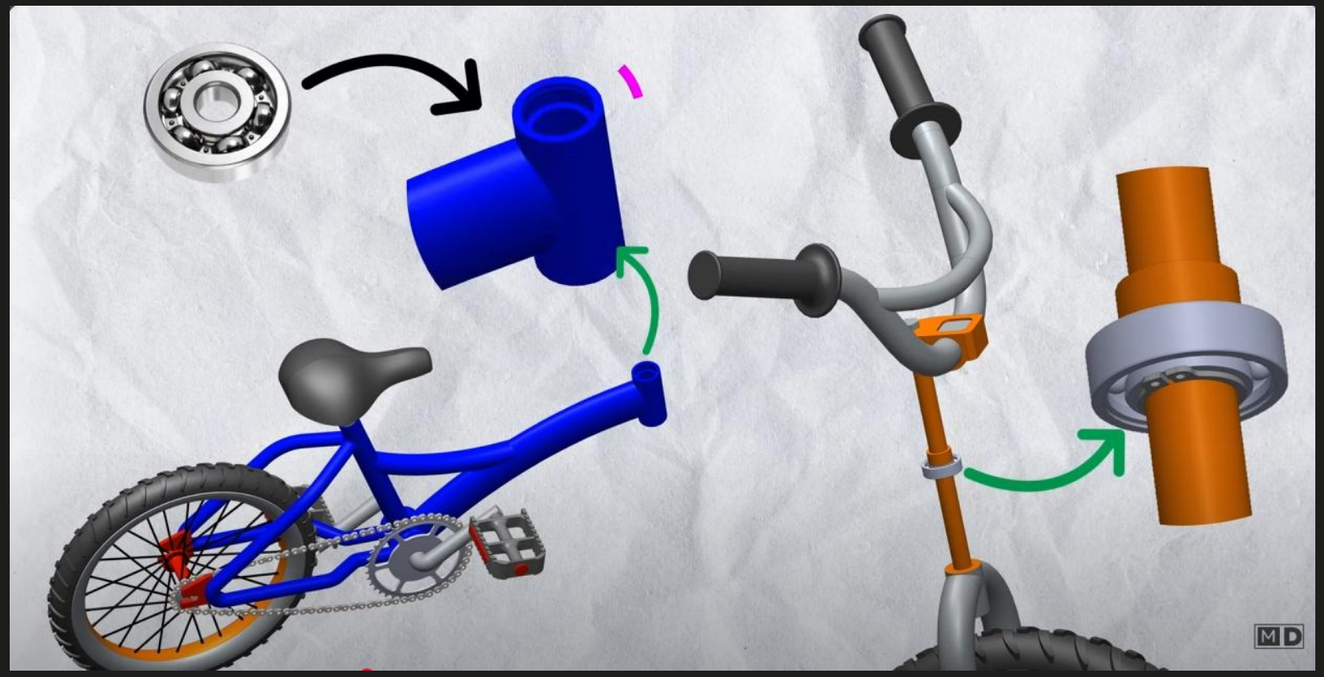
Needle Roller
Thrust Bearing

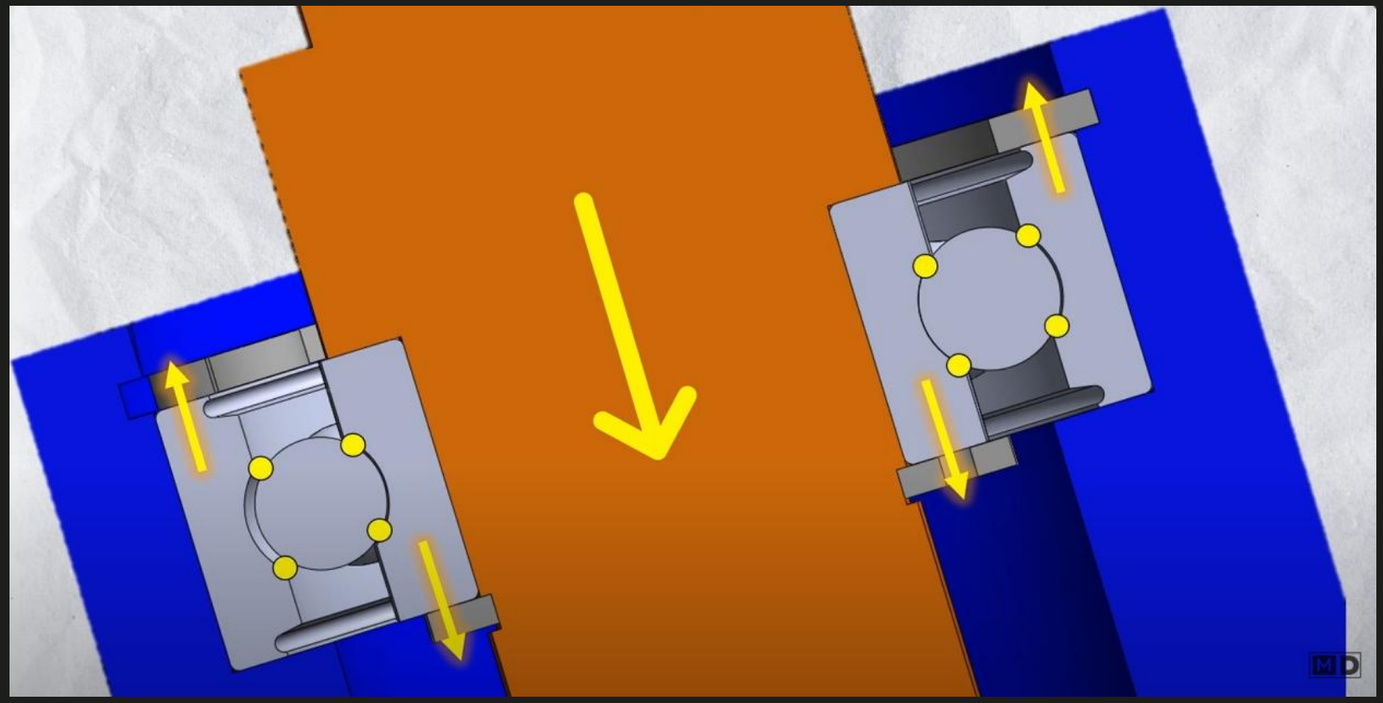
MD


Thrust Collar
Bearing
Bearing


Flat Land
Bearing
Tilting Pad
Bearing

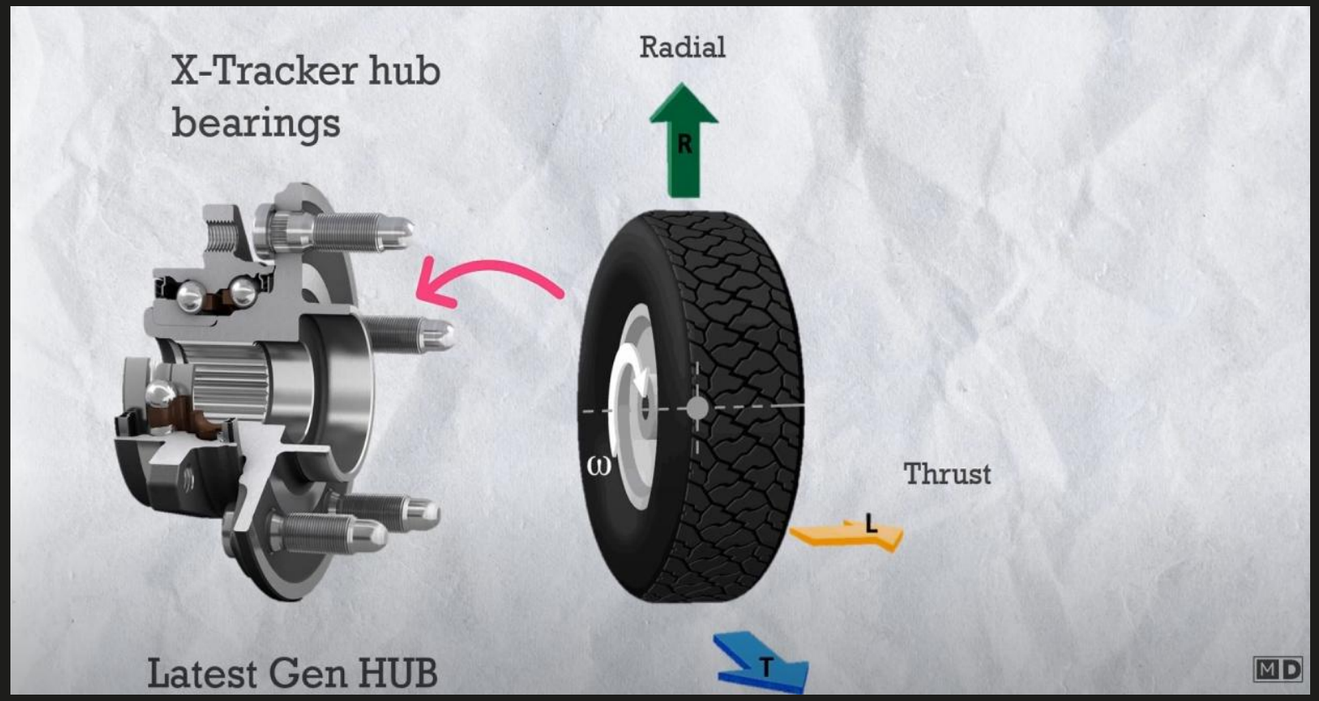











X-Tracker hub bearings
Radial
R
ω
Thrust
L
T
Latest Gen HUB



Linear Sliding Bearings

Linear sliding bearings are designed to facilitate linear motion between components. Common designs include bushings and guide ways that enable smooth and controlled linear movement.

Machine Tools

Precision movement for cutting and shaping.

Automated Assembly Lines

Efficient and reliable linear motion.

Linear Actuators

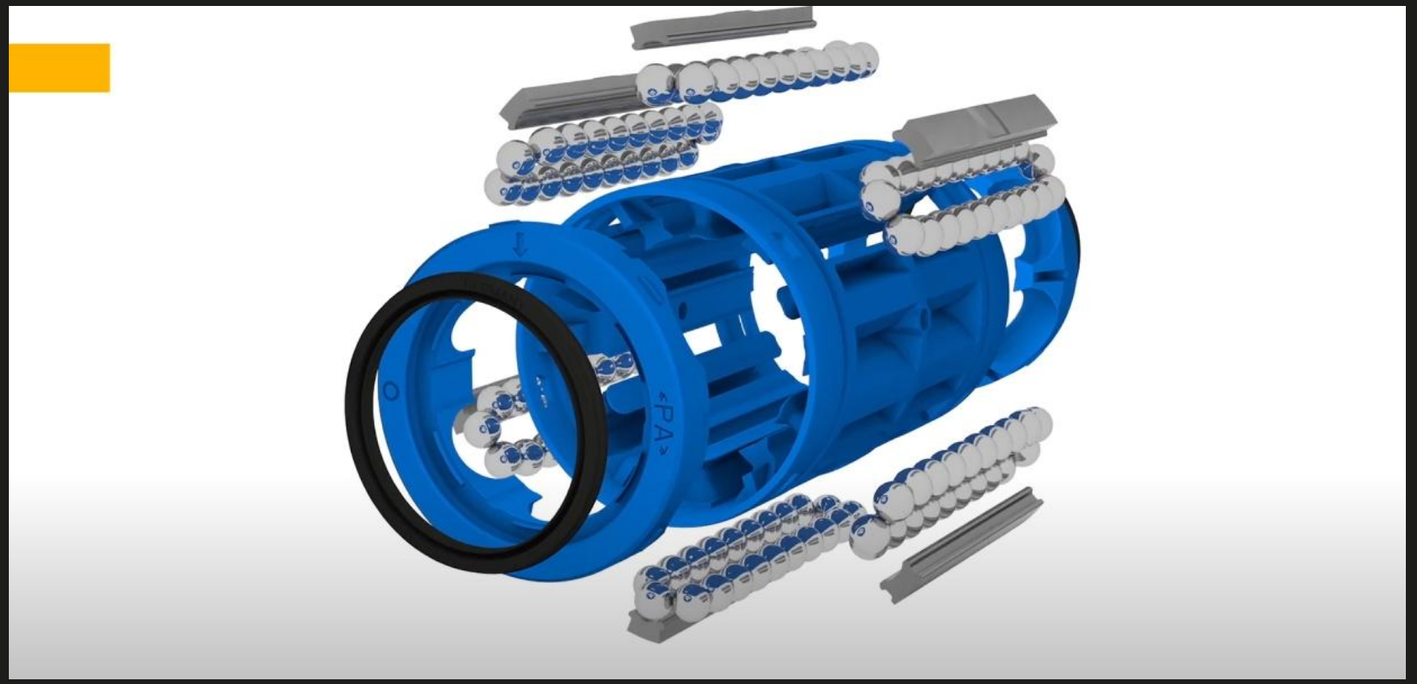
Controlled positioning in various applications.

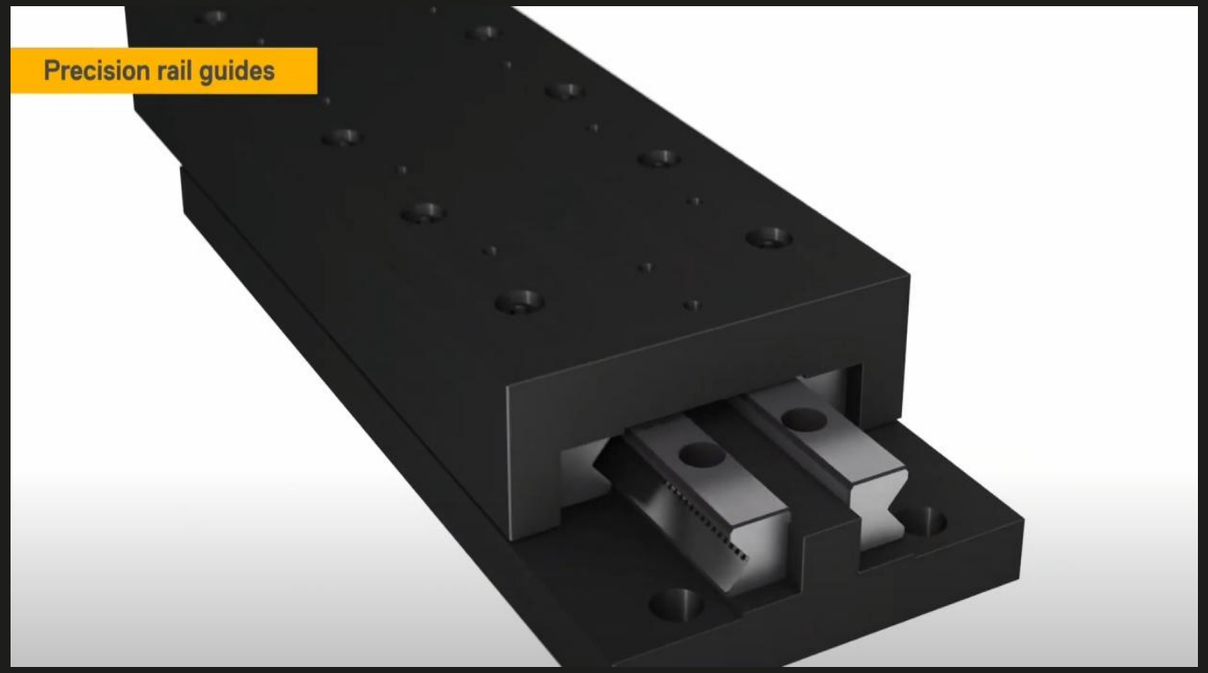

Gearbest
LM12UU High Precision / Stable Linear Motion Bearing


Gearbest


egov
महा
मे






Precision rail guides


Precision rail guides



Lubrication Regimes

The performance and lifespan of sliding contact bearings are heavily influenced by the lubrication regime in which they operate. Proper lubricant selection is critical, taking into account viscosity, additives, and operating temperature. Different regimes affect friction, wear, and efficiency.

1

Hydrodynamic Lubrication

Thick film, minimal wear, high efficiency.

2

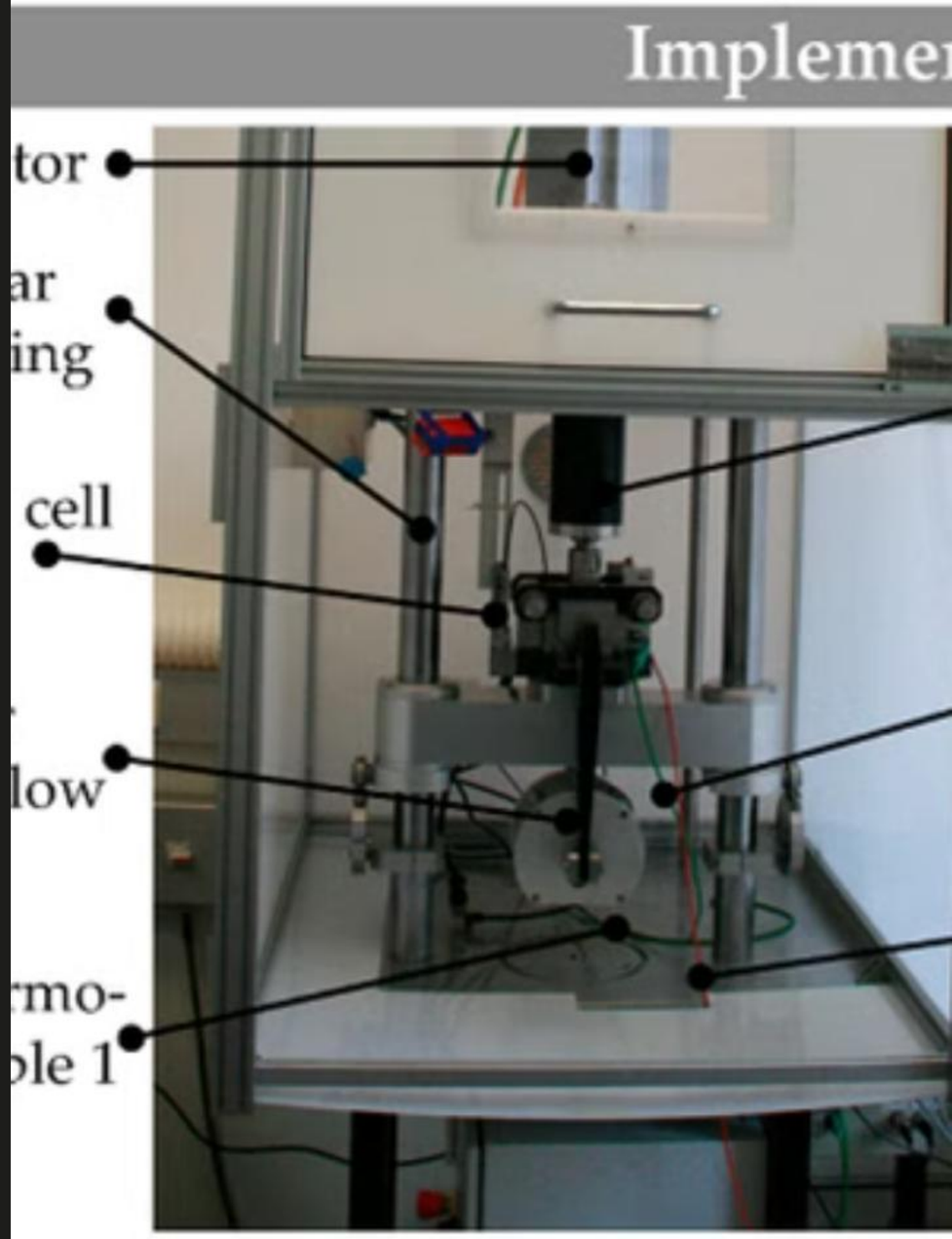
Mixed-Film Lubrication

Partial asperity contact, increased friction and wear.

3

Boundary Lubrication

Thin film, significant asperity contact, high friction and wear.



Materials for Sliding Contact Bearings

The materials used in sliding contact bearings play a crucial role in their performance and durability. Each material offers a unique combination of properties that make it suitable for specific applications.

Babbitt Alloys

Excellent conformability and embeddability, ideal for high-load applications (e.g., SnSb8Cu4 or PbSb15Sn10).

Bronze

High strength, corrosion resistance, and good wear properties (e.g., CuSn8, CuAl10Fe5Ni5).

Aluminum Alloys

High load capacity, good fatigue resistance, and lubricant compatibility (e.g., AlSn20Cu1).

Polymers

Low friction, self-lubricating, and lightweight, suitable for low-load applications (e.g., PTFE, PEEK, Nylon).



Advantages of Sliding Contact Bearings

Sliding contact bearings offer several advantages that make them suitable for a wide range of applications. These advantages include their high load capacity, compact size, damping capacity, and cost-effectiveness.



High load capacity

Can withstand significant static and dynamic loads

Compact size

Smaller footprint compared to rolling-element bearings

Damping capacity

Ability to absorb vibrations and reduce noise

Disadvantages of Sliding Contact Bearings

Despite their advantages, sliding contact bearings also have several drawbacks. These include higher friction, demanding lubrication requirements, susceptibility to wear, and sensitivity to contamination. Effective filtration systems and regular oil changes are critical to mitigate these issues.

1. ## Higher Friction

   Increased energy loss and heat generation.

2. ## Lubrication Requirements

   Need for continuous lubricant supply.

3. ## Wear

   Susceptible to wear under boundary lubrication.

4. ## Contamination Sensitivity

   Contaminants can accelerate wear.

Applications Across Industries

Sliding contact bearings find extensive applications across various industries due to their unique characteristics and capabilities. They are integral components in internal combustion engines, turbomachinery, pumps, and electric motors. The specific application depends on the bearing type and material.

Engines

Crankshaft and connecting rod bearings



Turbomachinery

Turbine and compressor bearings

Motors

Sleeve bearings in fractional horsepower motors

Pumps

Supporting shafts in centrifugal and positive displacement pumps

Future Trends and Conclusion

The field of sliding contact bearings is continually evolving with advancements in materials, the integration of smart technologies, and a growing focus on sustainable practices. These advancements aim to improve bearing performance, longevity, and environmental impact. Sliding contact bearings remain a vital component in numerous applications.

1

Advancements

New alloys and composites.

2

Smart Bearings

Sensors for monitoring condition.

3

Sustainable

Bio-based lubricants.

Reynolds's equation (2D):-

It describes how **pressure builds up** in a thin lubricant film between two surfaces in relative motion, such as in **journal bearings, thrust bearings, and slider bearings**.

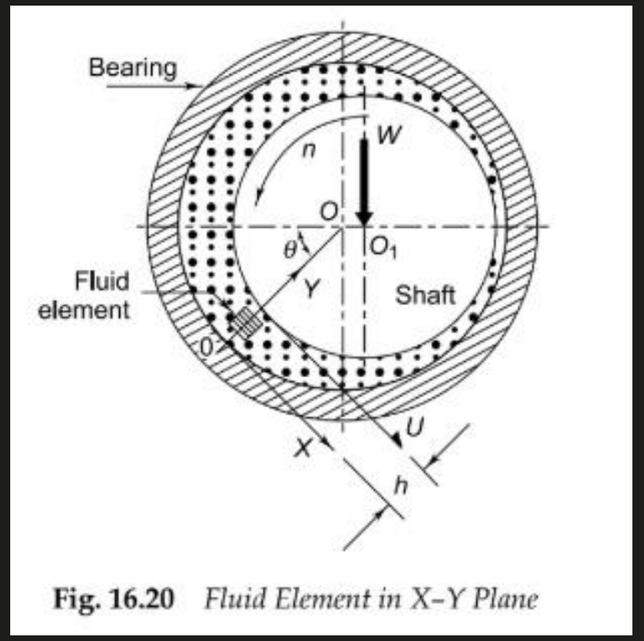


Fig. 16.20 *Fluid Element in X–Y Plane*

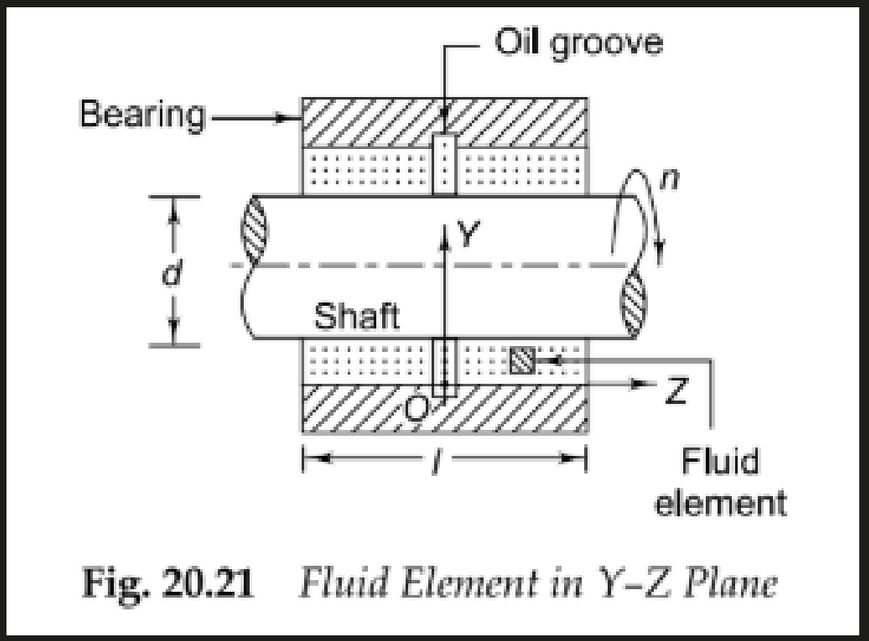


Fig. 20.21 *Fluid Element in Y–Z Plane*

$$\frac{\partial}{\partial x}\left[h^3\frac{\partial p}{\partial x}\right]+\frac{\partial}{\partial z}\left[h^3\frac{\partial p}{\partial z}\right]=6\mu U\left(\frac{\partial h}{\partial x}\right)$$

p → Pressure in the lubricant film.
h → Film thickness between the journal (shaft) and the bearing.
x, z → Spatial coordinates along the bearing surface
μ → Dynamic viscosity of the lubricant.
U → Surface velocity (usually the velocity of the moving shaft/journal).

(i) The lubricant obeys Newton's law of viscosity.

(ii) The lubricant is incompressible.

(iii) The inertia forces in the oil fi lm are negligible.

(iv) The viscosity of the lubricant is constant.

(v) The effect of curvature of the fi lm with respect to fi lm thickness is neglected. It is assumed that the fi lm is so thin that the pressure is constant across the fi lm thickness.

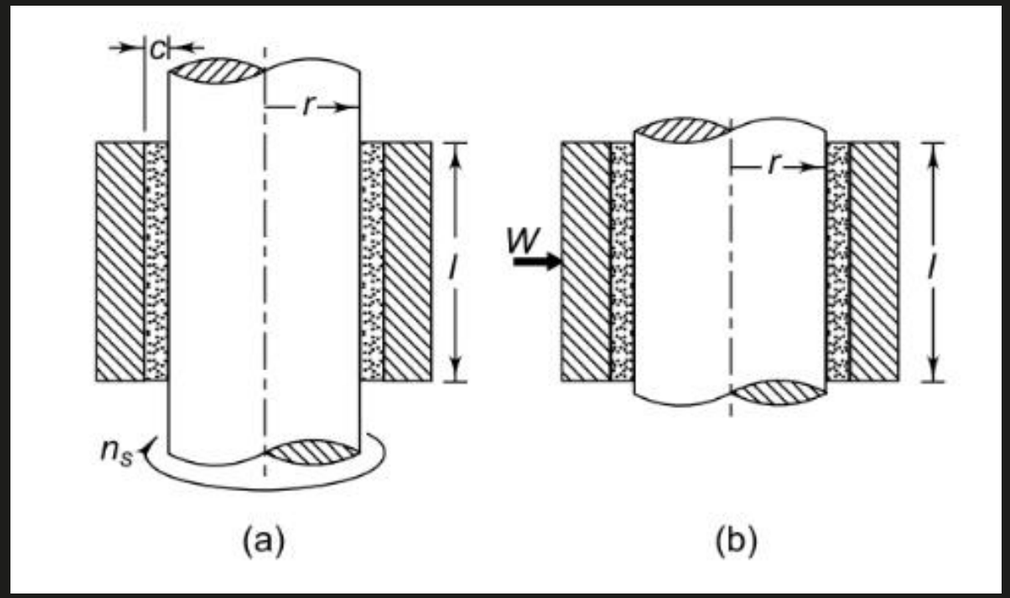
(vi) The shaft and the bearing are rigid.

(vii) There is a continuous supply of lubricant.

Petroff's equation:-

Petroff's equation is used to determine the coefficient of friction in journal bearings.

Assumptions:-
(i) The shaft is concentric with the bearing.
(ii) The bearing is subjected to light load.



$$f = (2\pi^2)\left(\frac{r}{c}\right)\left(\frac{\mu n_s}{p}\right)$$

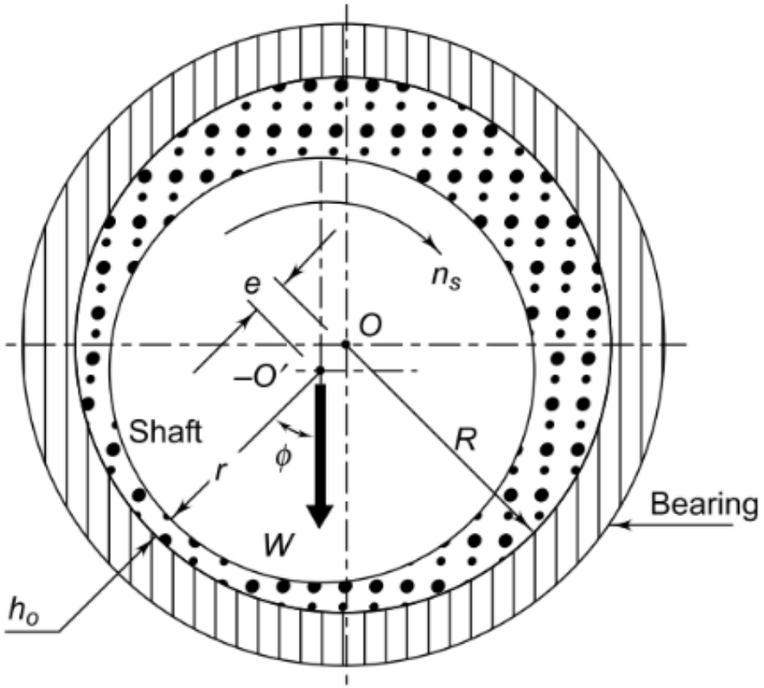
r = radius of the journal (mm)
l = length of the bearing (mm)
c = radial clearance (mm)
ns = journal speed (rev/sec)

Sommerfeld number:-

The **Sommerfeld number (S)** is a **dimensionless parameter** used in the design and analysis of **hydrodynamic journal bearings**. It represents the relationship between **bearing load, lubricant viscosity, journal speed, and clearance**.

$$S = \left(\frac{r}{c}\right)^2 \frac{\mu n_s}{p}$$

S = Sommerfeld number (dimensionless)
m = viscosity of the lubricant (N-s/mm2) or (MPa-s)
ns = journal speed (rev./s)
p = unit bearing pressure, i.e., load per unit of the projected area (N/mm2)



Sommerfeld Number (S)	Lubrication Condition	Bearing Performance
$S \gg 1$ **(Large S)**	Full hydrodynamic lubrication	Stable operation, no contact
$S \approx 1$ **(Moderate S)**	Transition zone	Risk of mixed lubrication
$S \ll 1$ **(Small S)**	Boundary lubrication	Direct contact, increased wear

Parameters of bearing design:-

1. Load Carrying Capacity (W)

- The bearing must **support the applied load** without excessive deformation or failure.
- **Equation:**

$$W = P \times A$$

where:

- W = Load (N)
- P = Bearing pressure (N/m^2)
- A = Load-bearing area (m^2)

- **Design Consideration:**
  - Bearings for **high-load applications** (e.g., turbines, marine engines) use materials with high **compressive strength** (e.g., bronze, babbitt).

2. Bearing Pressure (P)

- **Defines the stress on the bearing surface due to the applied load.**
- Excessive pressure can lead to **material deformation and wear.**

$$P = \frac{W}{L \times D}$$

where:

- L = Bearing length (m)
- D = Journal diameter (m)
- **Optimal Range:** Typically **0.5 to 2.5 MPa** for hydrodynamic bearings.

3. Clearance (C)

- **The radial gap between the journal and the bearing surface.**
- Ensures proper lubricant flow and minimizes friction.

$$C = R - r$$

where:

- R = Bearing radius (m)
- r = Shaft radius (m)
- **Typical clearance values:**
  - **Hydrodynamic bearings:** $0.001D$ to $0.002D$
  - **Precision applications:** $0.0005D$

4. Lubricant Viscosity (μ)

- Determines the **fluid film thickness** and **frictional losses**.
- Higher viscosity improves film strength but increases resistance.
- **Viscosity Selection:**
  - **Low-speed, high-load:** Use high-viscosity oil.
  - **High-speed applications:** Use low-viscosity oil.
- **Example:**
  - **SAE 30-50 oil** for IC engine crankshaft bearings.
  - **ISO VG 32-68 oil** for turbines and compressors.

5. Minimum Oil Film Thickness (h_{min})

- Ensures **no metal-to-metal contact** occurs.
- Depends on **load, speed, viscosity, and clearance.**

$$h_{min} = C \times \left(\frac{\mu N}{P} \right)$$

- **If h_{min} is too small → Bearing wear and failure.**
- **If h_{min} is too large → Excessive friction and energy loss.**

6. Bearing Friction (f)

- Depends on **shear resistance of the lubricant.**
- Affects **efficiency and energy loss** in rotating systems.
- **Friction Coefficient Equation:**

$$f = \frac{2\pi \mu N R^2}{W}$$

where:

- N = Rotational speed (rev/s)
- R = Journal radius
- **Typical friction coefficients:**
  - Hydrodynamic Bearings: **0.001 to 0.005**
  - Boundary Lubrication: **0.05 to 0.1**

7. Bearing Temperature Rise (ΔT)

- Heat is generated due to **viscous friction**.
- Excessive temperature can cause **lubricant breakdown**.
- **Heat Generation Equation:**

$$Q = \frac{fWV}{\eta}$$

where:

- Q = Heat generated (W)
- V = Sliding velocity (m/s)
- **Cooling Methods:**
  - **Forced lubrication** (external oil supply).
  - **Oil coolers** to maintain temperature.

8. Sommerfeld Number (S)

- A dimensionless parameter that determines lubrication conditions.

$$S = \frac{(r/c)^2 \cdot \mu N}{P}$$

- $S > 1$: Hydrodynamic lubrication (Stable operation).
- $S < 1$: Boundary lubrication (Risk of wear and failure).

9. Aspect Ratio (L/D)

- Defines the shape of the bearing.
- Affects pressure distribution and lubrication.
- Typical values:
  - Short bearings: $L/D < 1$ (Low friction, low stability).
  - Long bearings: $L/D > 1$ (Higher load capacity, risk of edge loading).

Conclusion

A properly designed bearing must balance **load, clearance, viscosity, and lubrication** to ensure **efficient operation, minimal wear, and long service life**.

THANK-YOU

